

| Question Number | Answer                                                                                                                                                                                                                                                                                                                             | Additional guidance | Mark       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------|
| <b>4(a)(i)</b>  | <p>A description that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>the germination of the seeds remains {high/ constant} for an initial period of time (1)</li> <li>(after this) the germination of the seeds (rapidly) {decreases / reduces} with increased time in storage (1)</li> </ul> |                     | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Additional guidance                                                                                                                                                                                                                                                                  | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>4(a)(ii)</b> | <p>An explanation that makes reference to four of the following points:</p> <ul style="list-style-type: none"> <li>(many different plants result in) increased genetic diversity of the seeds stored (1)</li> <li>because more (seeds from different) {individuals/ plants} increases the probability of having {more alleles / different genotypes / heterozygotes} (1)</li> <li>adult plants allow {pollination / sexual reproduction} to occur (1)</li> <li>to produce new seeds (for storage) (1)</li> <li>to replace those seeds stored for a long time (1)</li> </ul> | <p>Accept reduces chance of seeds from (mainly) {homozygous recessive / diseased} plants</p> <p>Accept to produce seeds which are genetically different (from parent plants)</p> <p>Accept to increase the {viability of stored seeds / percentage of seeds that will germinate}</p> | <b>(4)</b> |

| Question Number | Answer                                                                                                                                                                                                              | Additional guidance                                                                                                                                                                                                                                      | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>4(b)(i)</b>  | <p>A calculation in which:</p> <ul style="list-style-type: none"> <li>• mass of cutlery per month calculated (1)</li> <li>• mass of cutlery per year calculated to two significant figures with unit (1)</li> </ul> | <p><u>Example of calculation</u></p> <p><math>(130000 \div 100) \times 60 = 78000 \text{ (kg)}</math></p> <p><math>(78000 \times 12) = 940000 \text{ kg} / 9.4 \times 10^5 \text{ kg}</math></p> <p>Correct answer with no working scores full marks</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                | Additional guidance                                                                                                                                         | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>4(b)(ii)</b> | <p>An answer that includes the following points:</p> <ul style="list-style-type: none"> <li>• more (cutlery) can (constantly) be made from avocado seeds (1)</li> <li>• as avocados are {continually being grown / can be grown by future generations} (1)</li> </ul> | <p>Accept carbon neutral / lower carbon emissions (than oil-based plastic production) / biodegradable</p> <p>Accept as avocado seeds are waste products</p> | <b>(2)</b> |